CSE 3203
Operating System and System Programming
Chapter 1: Introduction

Operating System

Objectives

- To describe the basic organization of computer systems
- To provide a grand tour of the major components of operating systems
- To give an overview of the many types of computing environments
- To explore several open-source operating systems

Chapter 1: Introduction

- **What Operating Systems Do**
- **Computer-System Organization**
- **Computer-System Architecture**
- **Operating-System Structure**
- **Operating-System Operations**
- **Process Management**
- **Memory Management**
- **Storage Management**
- **Protection and Security**
- **Kernel Data Structures**
- **Computing Environments**
- **Open-Source Operating Systems**

What is an Operating System?

- **A program that acts as an intermediary between a user of a computer and the computer hardware.**

- **Operating System is a software that controls the operation of a computer and directs the processing of programs (as by assigning storage space in memory and controlling input and output functions)**

Operating system goals:

- Execute user programs and make solving user problems easier
- Make the computer system convenient to use
- Use the computer hardware in an efficient manner

Computer System Structure

- **Computer system can be divided into four components:**
  - **Hardware** – provides basic computing resources
    - CPU, memory, I/O devices
  - **Operating system**
    - Controls and coordinates use of hardware among various applications and users
  - **Application programs** – define the ways in which the system resources are used to solve the computing problems of the users
    - Word processors, compilers, web browsers, database systems, video games
  - **Users**
    - People, machines, other computers

Four Components of a Computer System

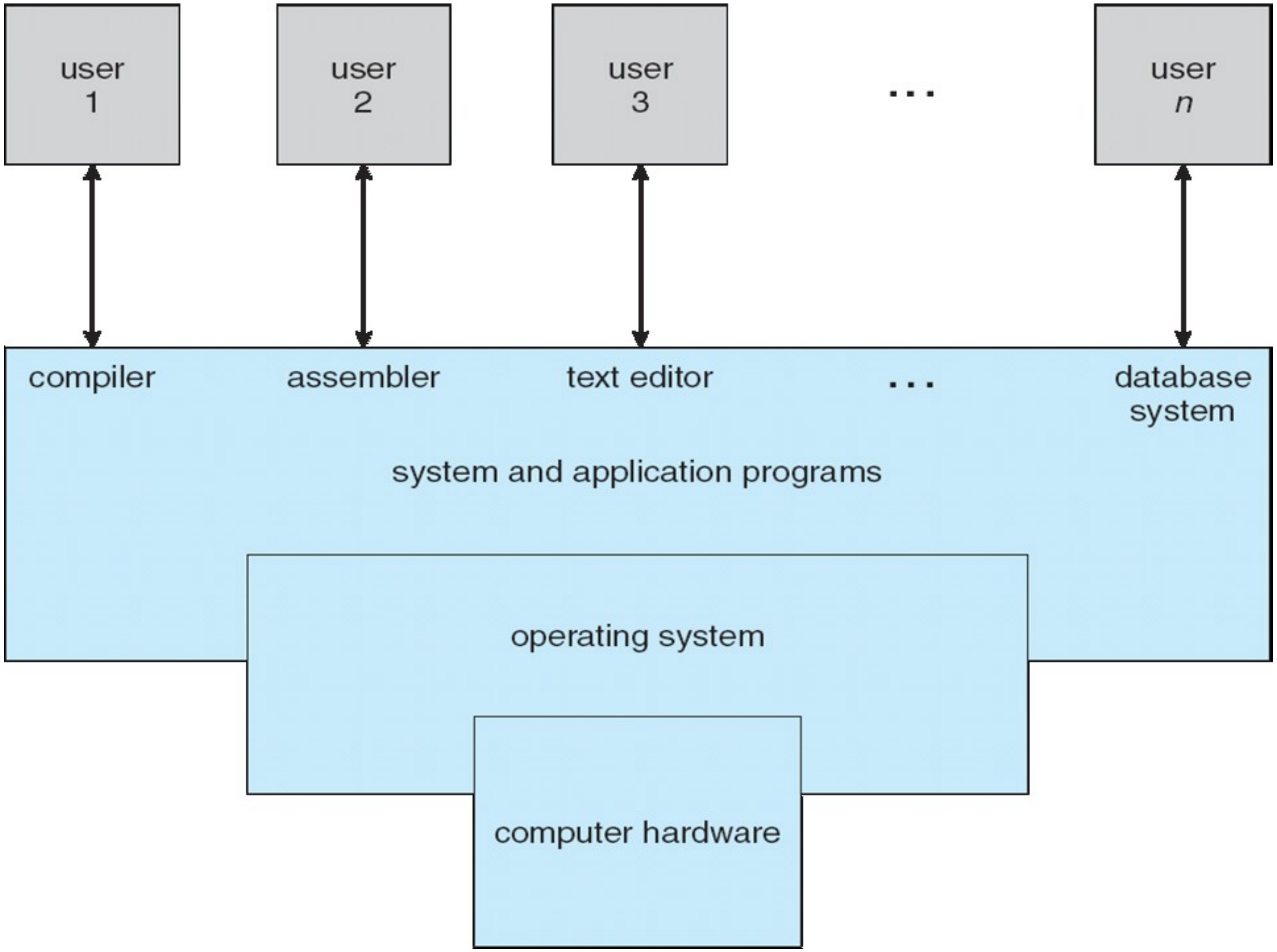


Fig. Abstract view of the components of a computer system.

What Operating Systems Do

- **Depends on the point of view**
- Users want convenience, **ease of use** and **good performance**
  - Don't care about **resource utilization**
- But shared computer such as **mainframe** or **minicomputer** must keep all users happy
- Users of dedicate systems such as **workstations** have dedicated resources but frequently use shared resources from **servers-**designed to compromise between individual usability and resource utilization.
- **Handheld computers** are resource poor, optimized for usability and battery life
- Some computers have little or no user interface, such as **embedded computers in devices and automobiles**

Operating System Definition

- OS is a **resource allocator**
  - Manages all resources
  - Decides between conflicting requests for efficient and fair resource use

- OS is a **control program**
  - Controls execution of programs to prevent errors and improper use of the computer

Operating System Definition (Cont.)

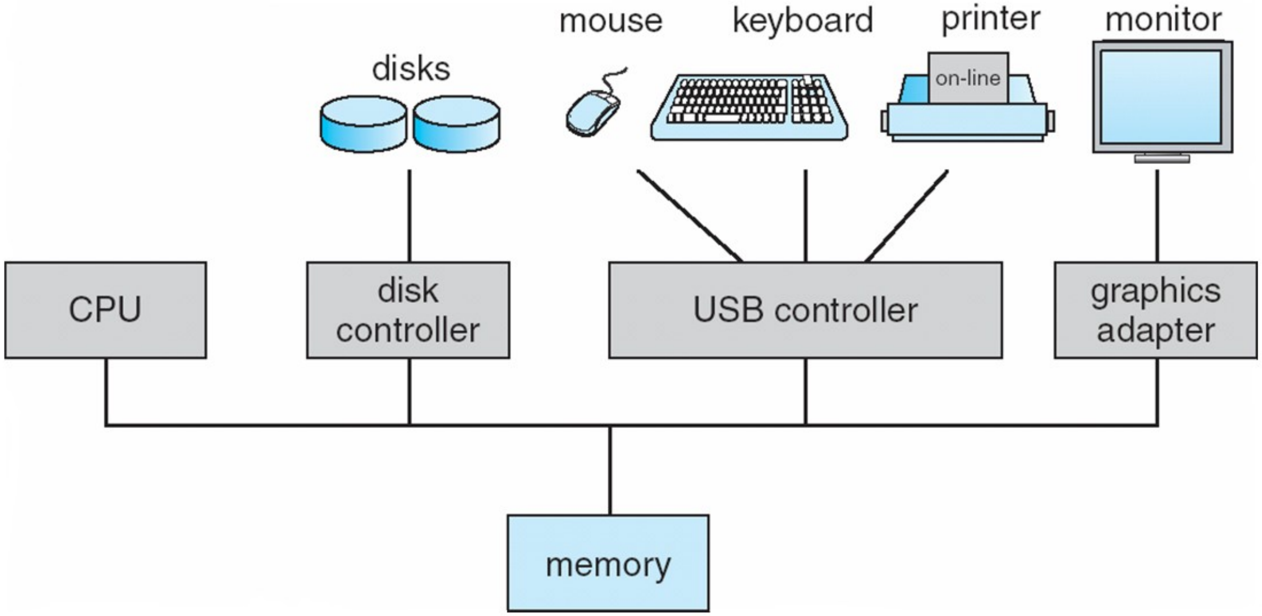
- “**The one program running at all times on the computer” is the kernel.**
- **Kernel: is the collection of all core software of operating system.**
- **middleware—a set of software frameworks that provide additional services to application developers**
- (Apple’s iOS and Google’s Android-features
- a core kernel along with middleware that supports databases, multimedia, and graphics)

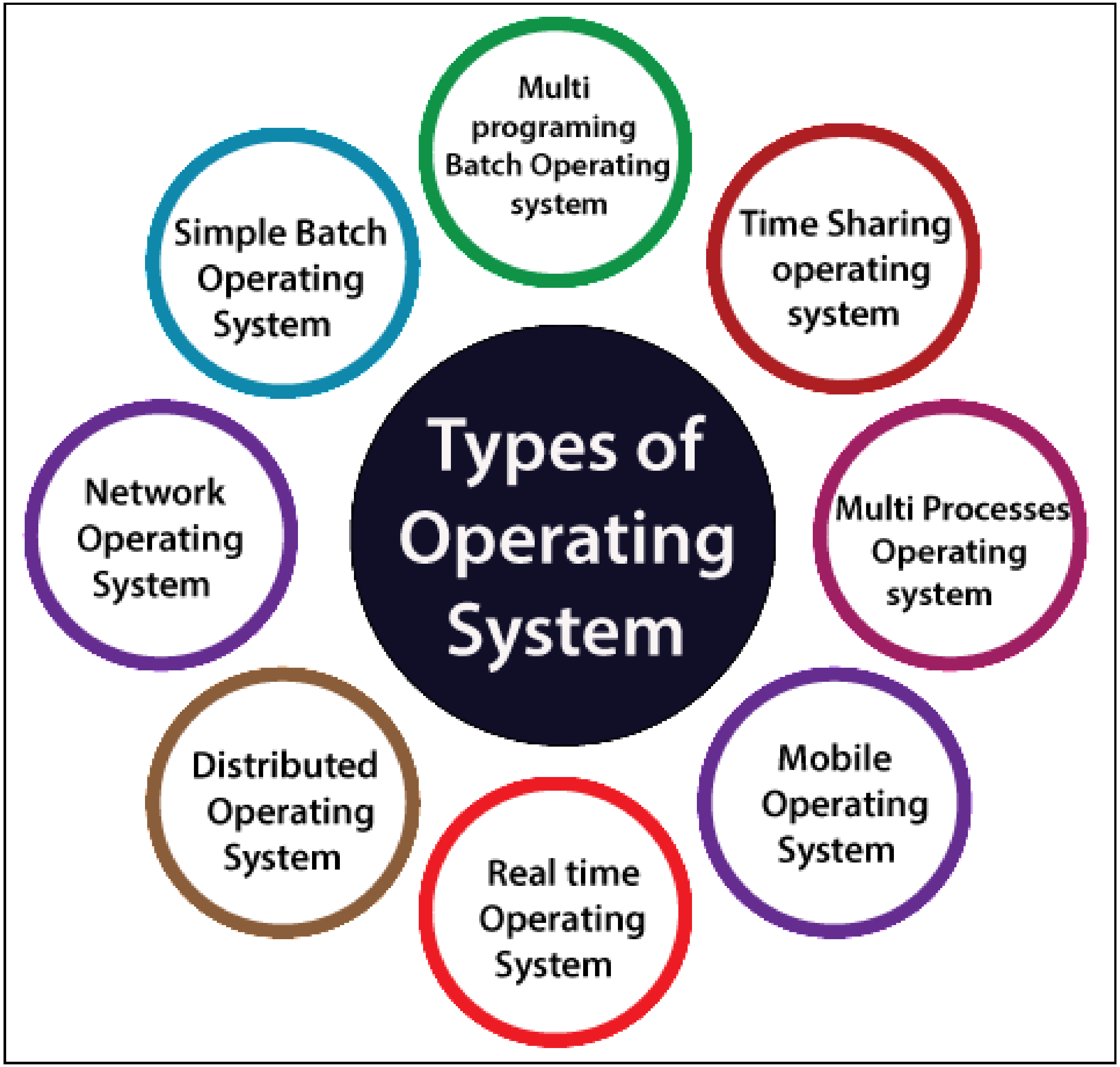
Bootstrap program

- **bootstrap or bootloader program** is loaded at power-up or reboot
  - Typically stored in ROM or EPROM, generally known as **firmware**
  - Initializes all aspects of system
  - Loads operating system kernel and starts execution

Computer System Organization

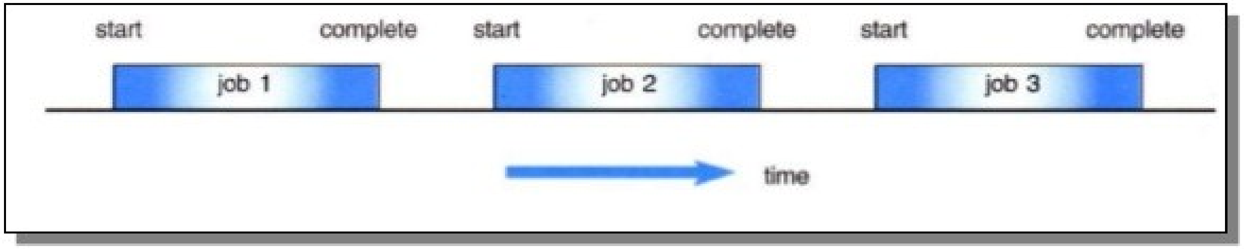
- Computer-system operation
  - One or more CPUs, device controllers connect through common bus providing access to shared memory
  - Concurrent execution of CPUs and devices competing for memory cycles




Types of Operating System
Multi programing Batch Operating system
Time Sharing operating system
Multi Processes Operating system
Mobile Operating System
Real time Operating System
Distributed Operating System
Network Operating System
Simple Batch Operating System

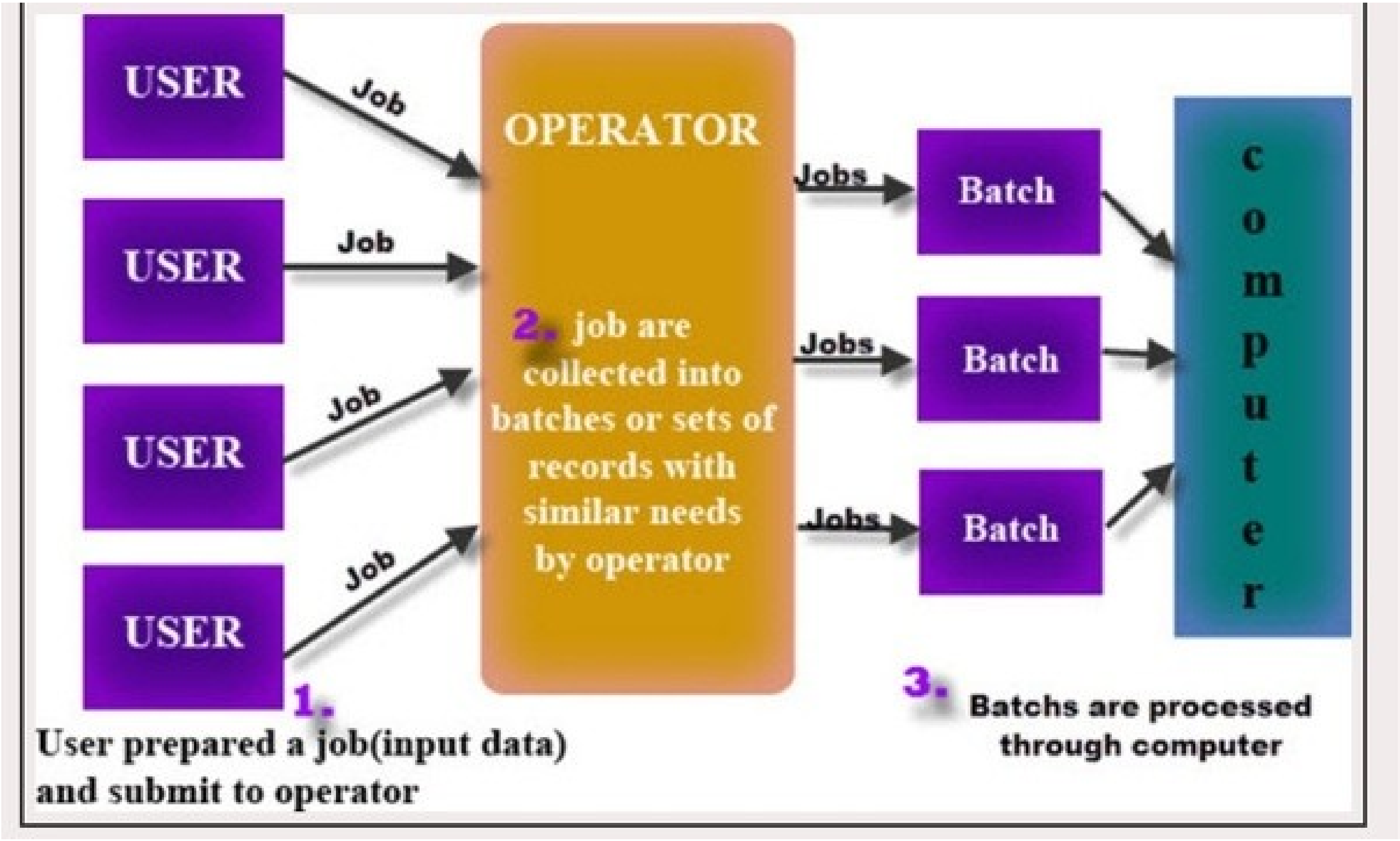
Batch operating systems

- **Batch Operating System:** **Batch processing** is the execution of non-interactive processing tasks, meaning tasks with no user-interface. To speed up processing, jobs with similar needs are batched together and run as a grou



- **How it works:** OS keeps the number of jobs in memory and executes them one by one. Jobs processed in first come first served order. Each set of a job considered as a batch. When a job completes its execution, its memory is released, and the output for the job gets copied into an output spool for later printing or processing.

Batch operating systems



Batch operating systems

- **Example of Batch Operating System:**
  - **Billing:**
- **A telecom company runs a monthly batch job to process call data records that include the details of millions of phone calls to calculate charges**
  - **Transactions**
- **A bank that processes transactions such as international money transfers after-hours. After-hours batch processing was once extremely common in the banking industry.**
- **Examples of Batch based Operating System:** Payroll System, Bank Statements etc

Types of operating systems

- **Batch Operating System:**

Advantages:

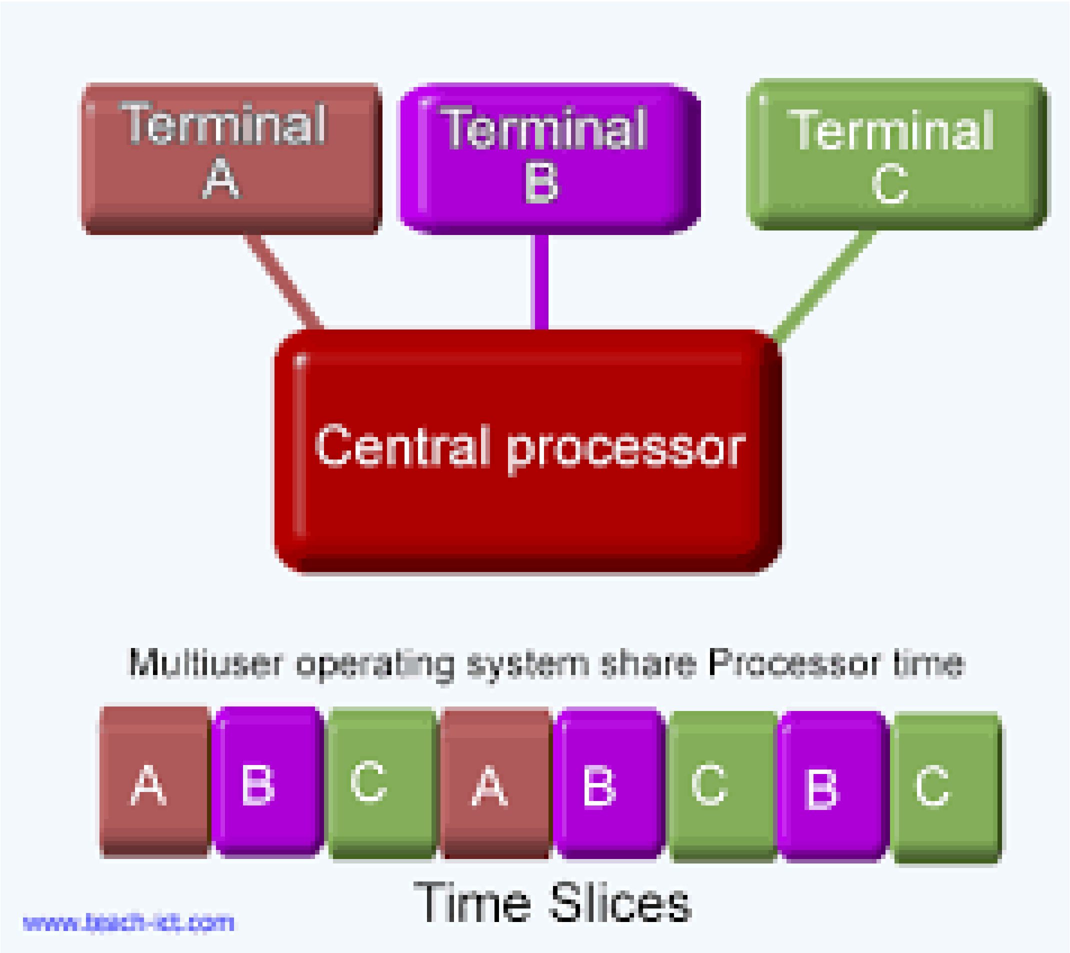
1) Same jobs in the batch are higher executed speed.
2) A process is complete its execution, next job from job spool get executed without any user interaction.
3) CPU utilization gets improved.
4) To speed up the processing speed, the batch process can partition into the number of processes.

Disadvantages:

1) Difficult to debug.
2) If a job gets to enter in an infinite loop, other jobs wait for unknown time.
3) Batch systems are costly

Time-Sharing operating systems

- **Time-Sharing Operating Systems :**
- Time-sharing is a technique which enables many people, located at various terminals, to use a particular computer system at the same time.
- Each task is given equal time to execute which is known as time slice or quantum, so that all the tasks work smoothly. Each user gets time of CPU as they use single system. These systems are also known as Multitasking Systems.
- Time-sharing or multitasking is a logical extension of multiprogramming.



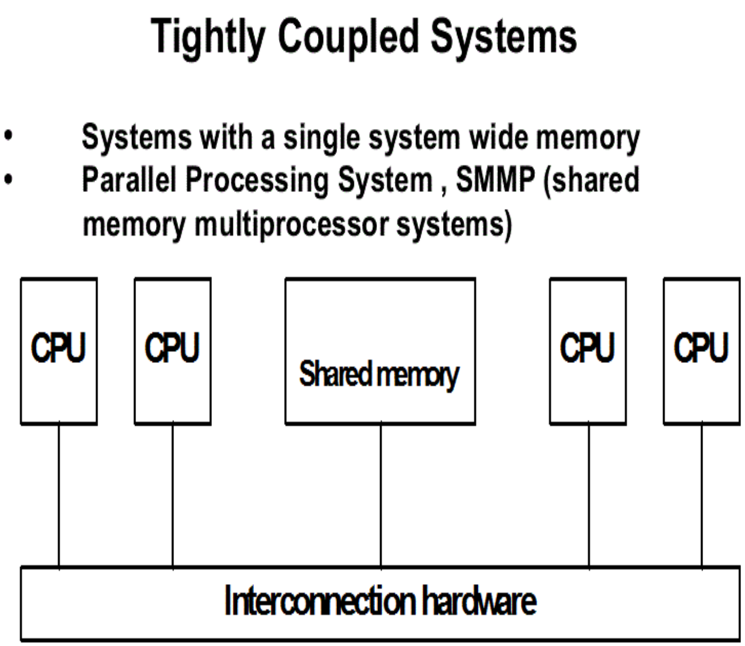
❖ **Examples of Time-Sharing OSs are:** Multics, Unix etc.

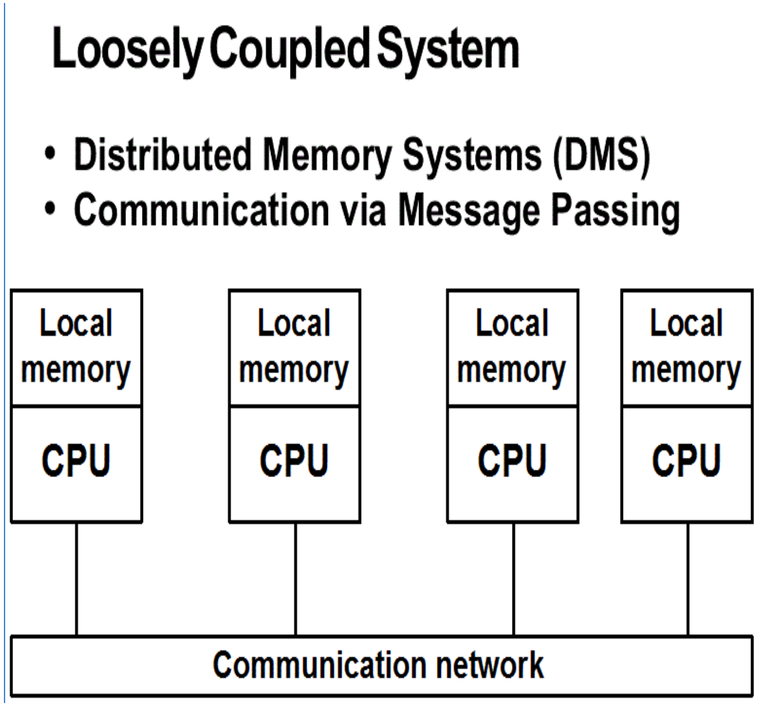
Distributed operating systems

- **Distributed Operating System:**
- A distributed system is a system whose components are located on different networked computers, which communicate and coordinate their actions by passing messages to one another.
- Various autonomous interconnected computers communicate each other using a shared communication network.
- Independent systems possess their own memory unit and CPU. These are referred as **loosely coupled systems** or distributed systems.

Loosely Coupled System vs. Tightly Coupled System

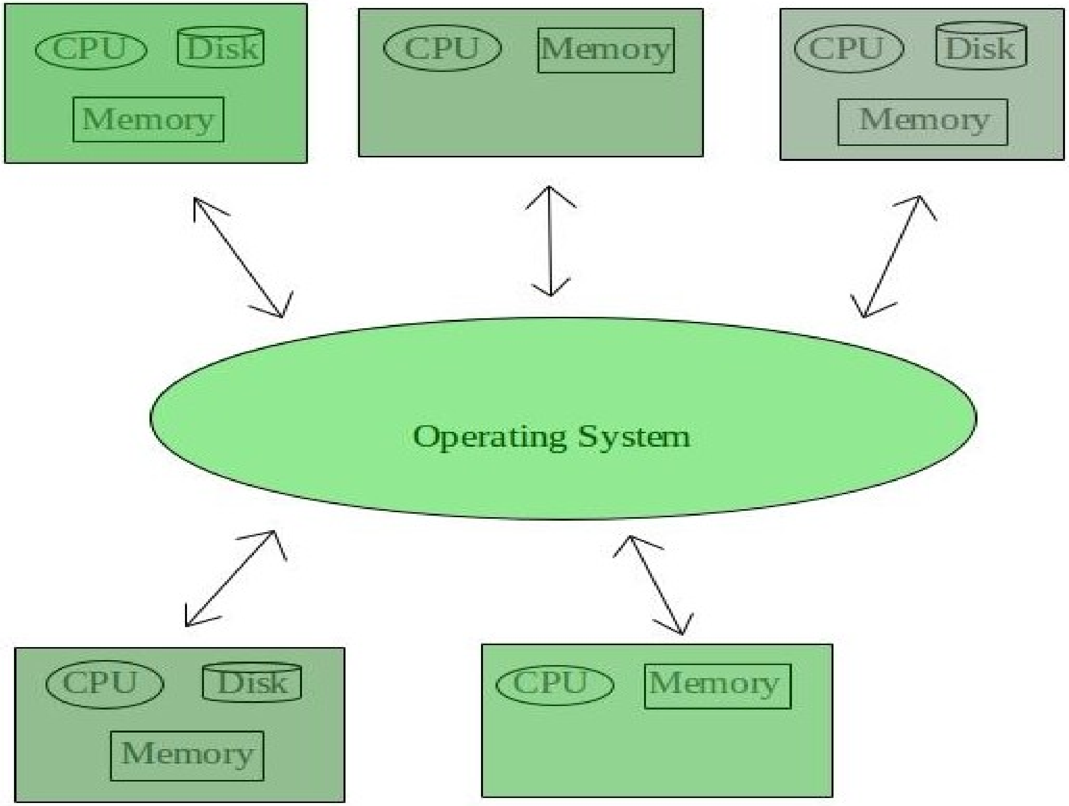
- A tightly coupled system consists of processors or modules that share the same memory and communicate directly through a high-speed bus.
- A loosely coupled system consists of independent processors or computers, each having its own memory and operating system. They communicate by exchanging messages over a network.





Distributed operating systems

- **Distributed Operating System:**



Examples of Distributed Operating System are- LOCUS, AEGIS etc.

Network Operating System

- **Network Operating System :**

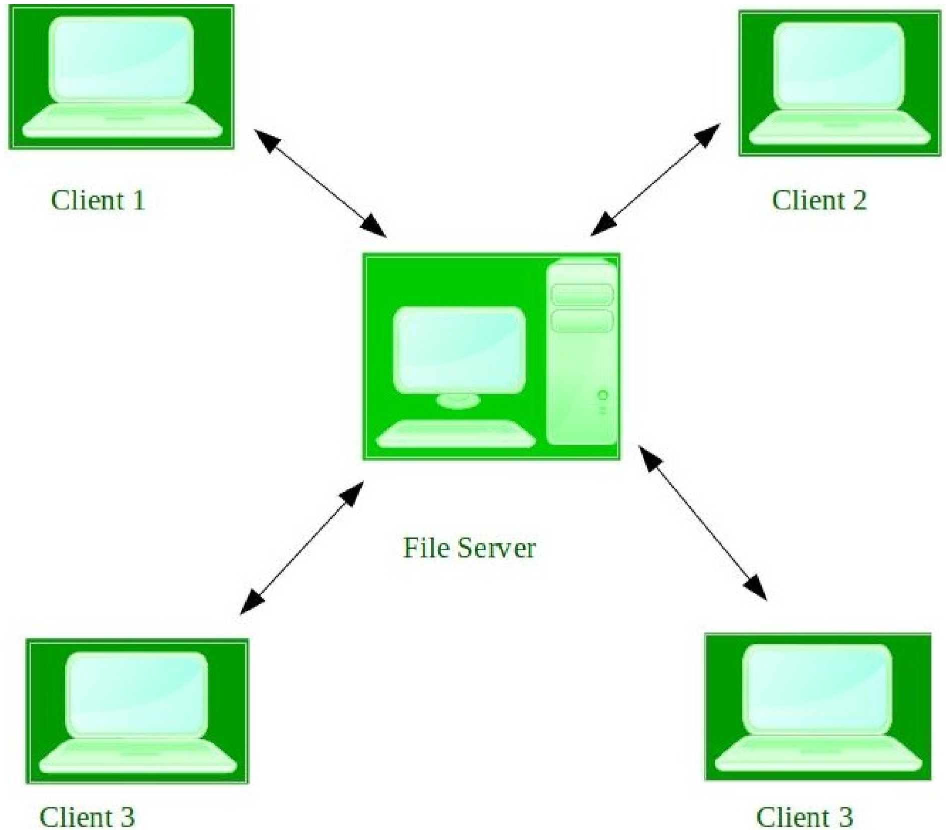
  an alternative explanation of an operating system entering the computer network that allows the sharing of content from one device to the other.

- These type of operating systems allow shared access of files, printers, security, applications, and other networking functions over a small private network.

Feature	Distributed OS	Network OS
Definition	Multiple computers work together as one system.	Multiple independent computers connected by a network.
User View	Appears as a single computer.	Users know they are using different computers.
System Image	Single System Image (SSI).	No single system image.
Transparency	High; location of resources is hidden.	Low; users explicitly access remote resources.
Operating System	Cooperating distributed OS across nodes.	Each computer runs its own OS independently.
Scalability	Excellent.	Good, but more manual administration is needed.
Administration	Appears centralized.	Each machine is administered separately.
Examples	Amoeba, Plan 9, Sprite	Windows Server, Ubuntu Server, UNIX Server

Types of operating systems

- **Network Operating System :**



Examples of Network Operating System are: Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X, Novell NetWare, and BSD etc.

Mobile operating systems

- Handheld smartphones, tablets, etc
- What is the functional difference between them and a "traditional" laptop?
- Extra feature – more OS features (GPS, gyroscope)
- Allows new types of apps like ***augmented reality***
- Use IEEE 802.11 wireless, or cellular data networks for connectivity
- Leaders are **Apple iOS** and **Google Android**

Embedded operating system

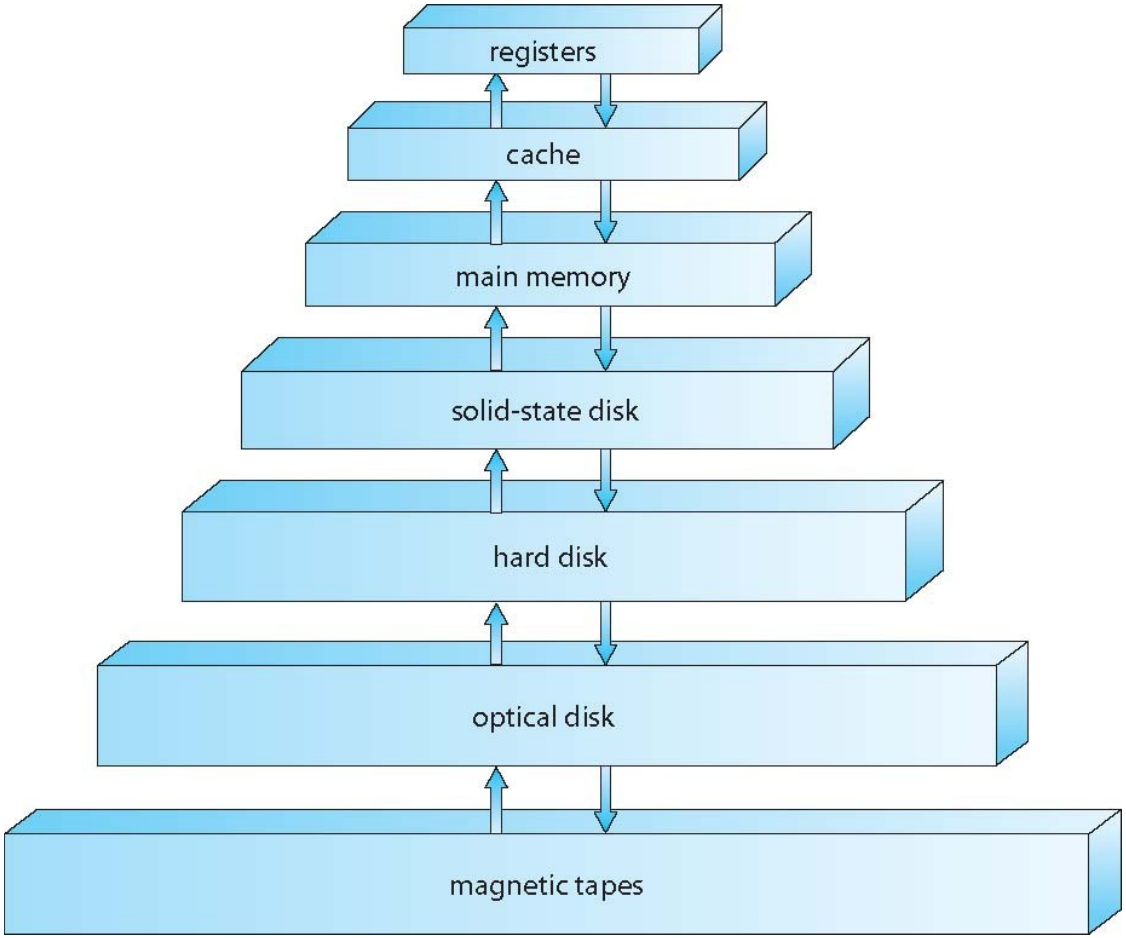
An **embedded operating system** is specialized for use in the computers built into larger systems, such as cars, traffic lights, digital televisions, ATMs, airplane controls, point of sale (POS) terminals, digital cameras, GPS navigation systems, elevators, digital media receivers and smart meters.

Types of operating systems

- A **real-time operating system (RTOS)** is an operating system that guarantees a certain capability within a specified time constraint. For example, an operating system might be designed to ensure that a certain object was available for a robot on an assembly line.

- **Examples of Real-Time Operating Systems are:**
- **Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.**

Storage-Device Hierarchy



Performance of Various Levels of Storage

Level	1	2	3	4	5
Name	registers	cache	main memory	solid state disk	magnetic disk
Typical size	< 1 KB	< 16MB	< 64GB	< 1 TB	< 10 TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time (ns)	0.25 - 0.5	0.5 - 25	80 - 250	25,000 - 50,000	5,000,000
Bandwidth (MB/sec)	20,000 - 100,000	5,000 - 10,000	1,000 - 5,000	500	20 - 150
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape

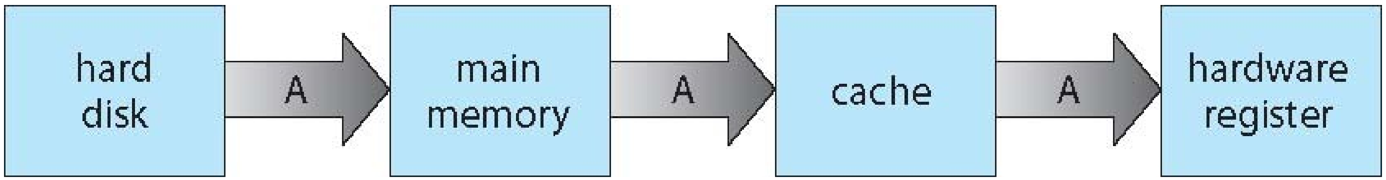
Movement between levels of storage hierarchy can be explicit or implicit

Caching

- Important principle, performed at many levels in a computer (in hardware, operating system, software)
- Information in use copied from slower to faster storage temporarily
- Faster storage (cache) checked first to determine if information is there
  - If it is, information used directly from the cache (fast)
  - If not, data copied to cache and used there

Migration of data "A" from Disk to Register

- Multitasking environments must be careful to use most recent value, no matter where it is stored in the storage hierarchy

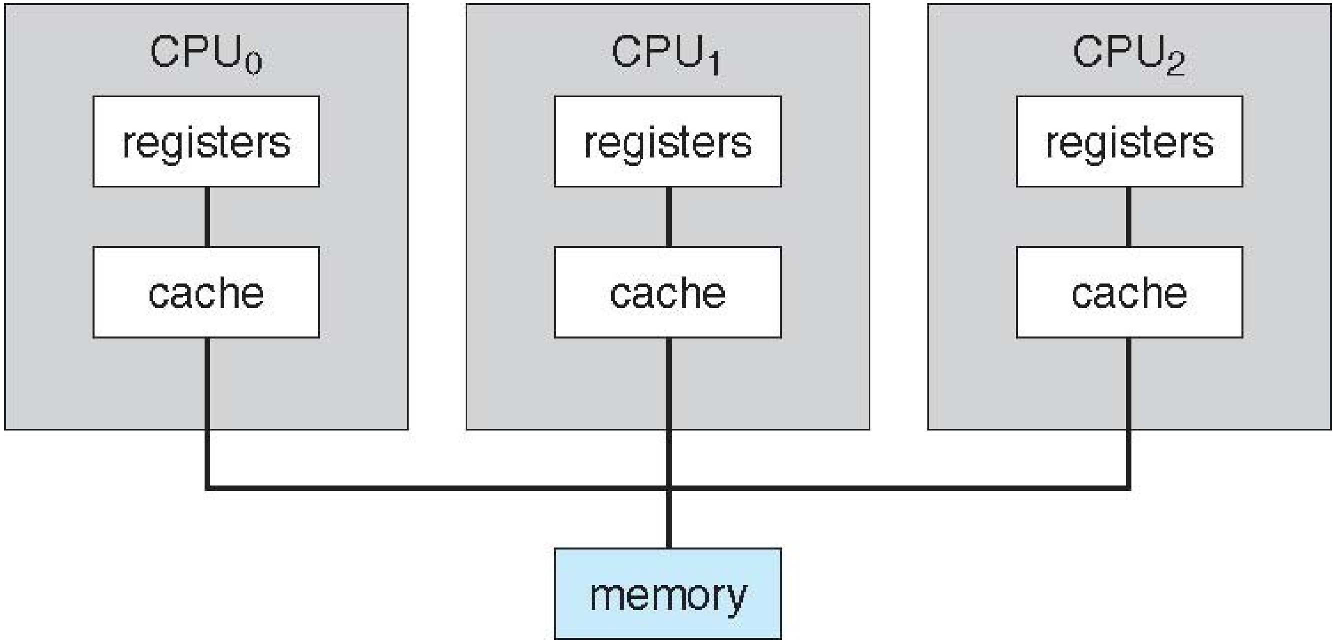


- Multiprocessor environment must provide **cache coherency** in hardware such that all CPUs have the most recent value in their cache

Computer-System Architecture

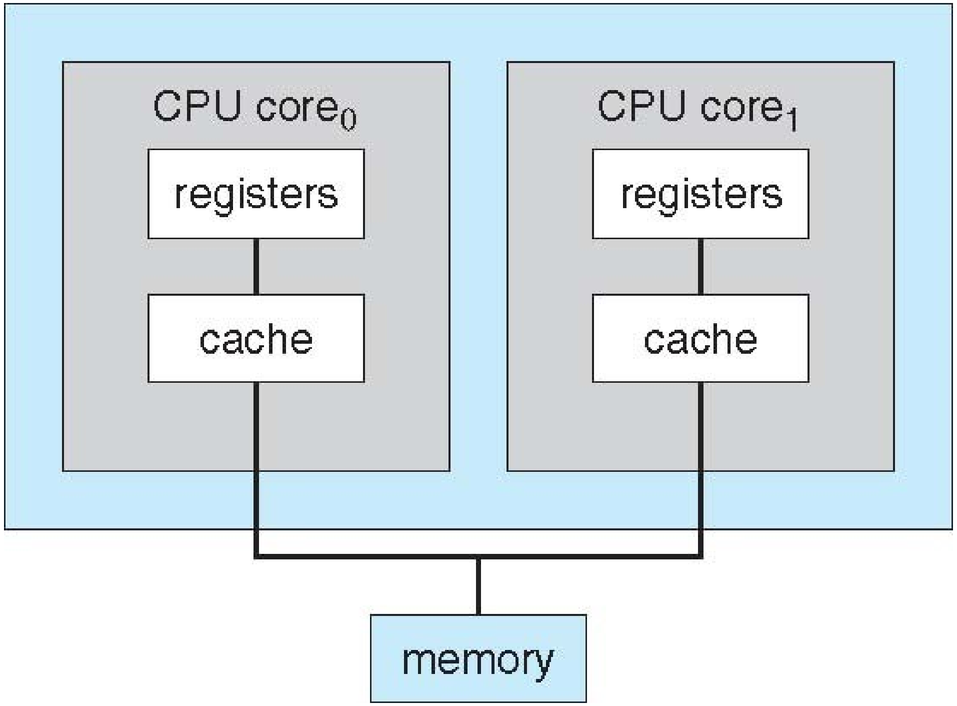
- Most systems use a single general-purpose processor
  - Most systems have special-purpose processors as well
- **Multiprocessors** systems growing in use and importance
  - Also known as **parallel systems**, **tightly-coupled systems**
  - Advantages include:
    1. **Increased throughput**
    2. **Economy of scale**
    3. **Increased reliability** – graceful degradation or fault tolerance
  - Two types:
    1. **Asymmetric Multiprocessing** – each processor is assigned a specific task.
    2. **Symmetric Multiprocessing** – each processor performs all tasks

Symmetric Multiprocessing Architecture

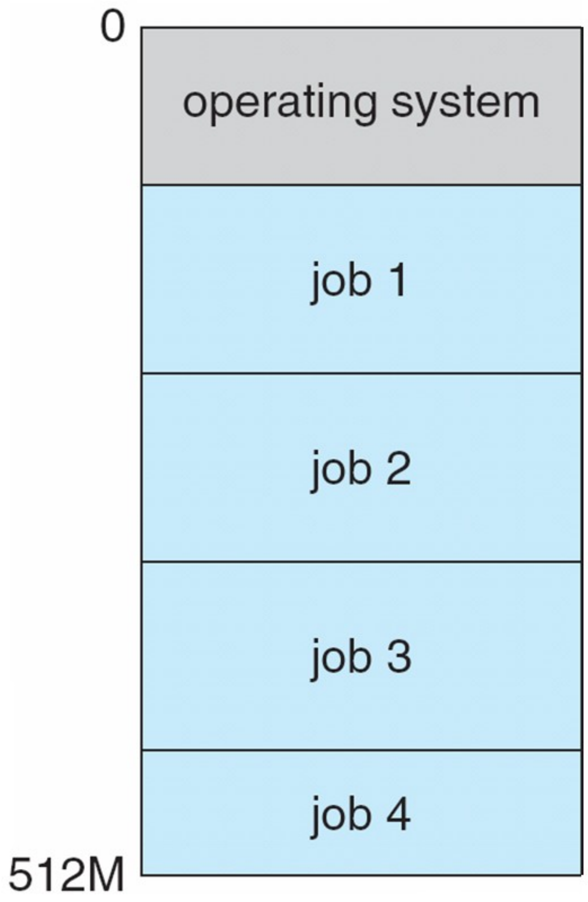


A Dual-Core Design

- Multi-chip and **multicore**
- Systems containing all chips
  - Chassis containing multiple separate systems



Memory Layout for Multiprogrammed System



Operating-System Operations

- **Dual-mode** operation allows OS to protect itself and other system component
  - **User mode** and **kernel mode**
  - **Mode bit** provided by hardware
    - Provides ability to distinguish when system is running user code or kernel code
    - Some instructions designated as **privileged**, only executable in kernel mode
    - System call changes mode to kernel, return from call resets it to user

Examples of Privileged Instructions:

- Context Switching
- Clearing Memory or Removing process from Memory
- Modify entries in Device-status table
- I/O instructions and Halt instructions
- Turn off all Interrupts
- Setting Timer
- **Examples of Non-Privileged Instructions:**
- Reading the status of Processor
- Reading the System Time
- Generate any Trap Instruction (software interrupt)
- Sending the final printout of Printer

Transition from User to Kernel Mode

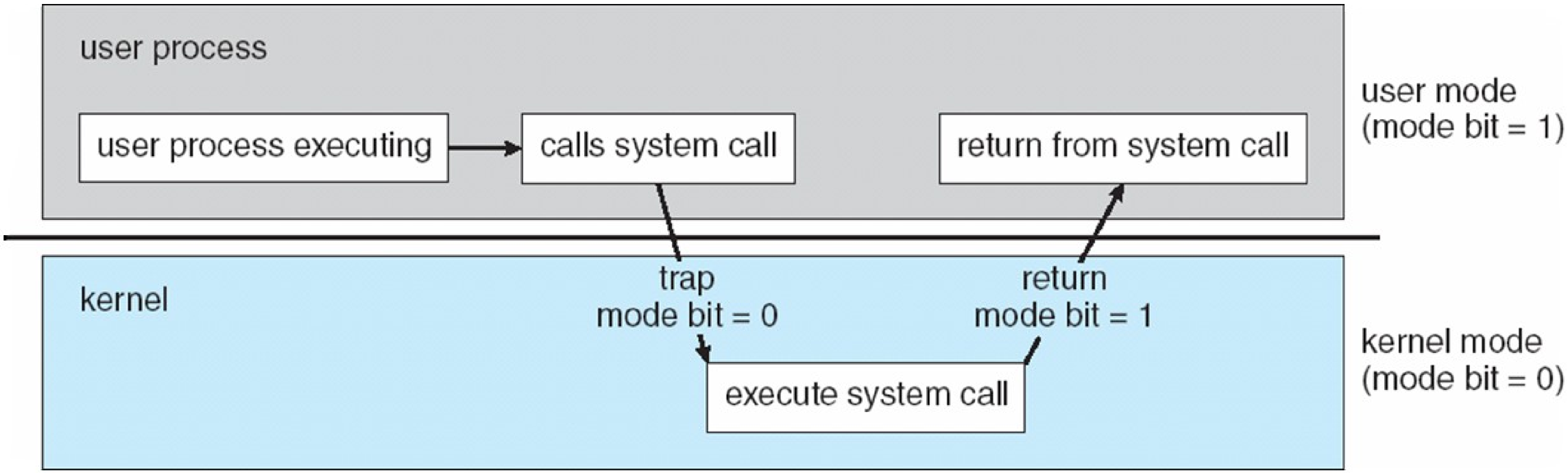


Figure: Dual mode operation

Open-Source Operating Systems

- Operating systems made available in source-code format rather than just binary **closed-source**
- **Linux is the best-known and most-used open source operating system.**
- Started by **Free Software Foundation (FSF)**, which has "copyleft" **GNU Public License (GPL)**
- https://opensource.com/resources/linux

Open-Source Operating Systems

- **Ubuntu**
- **Linux Lite**
- **Fedora**
- **Linux Mint**
- **Solus. Source**
- **Xubuntu**
- **Chrome OS**
- **React OS**
- (Source: https://blogs.systweak.com/8-best-open-source-operating-system/)

xubuntu

Linux Mint

from freedom came elegance

ReactOS

chromeOS

Computing Environments - Virtualization

- Allows operating systems to run applications within other OSes
  - Vast and growing industry
- Emulation used when source CPU type different from target type (i.e. PowerPC to Intel x86)
  - Generally slowest method
- Virtualization – OS natively compiled for CPU, running guest OSes also natively compiled
  - Consider VMware running WinXP guests, each running applications, all on native WinXP host OS
  - VMM (virtual machine Manager) provides virtualization services

Computing Environments - Virtualization